



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

CST 201-DATA STRUCTURES

1. What does the Big Ω notation represent?

- A) The upper bound of an algorithm's running time
- B) The lower bound of an algorithm's running time
- C) The average-case running time of an algorithm
- D) The worst-case running time of an algorithm

2. What is a data structure?

- a) A programming language
- b) A collection of algorithms
- c) A way to store and organize data
- d) A type of computer hardware

3. The prerequisite of the binary search algorithm is

- a) Array should be sorted in descending order
- b) Array should be randomly arranged
- c) Array should be sorted in ascending order
- d) None of these

4. State True or False

- i) Binary search is used for searching in a sorted array
 - ii) The time complexity of binary search is $O(\log n)$
- A) True, True
 - B) False, True
 - C) True, false
 - D) False, False

5. What is the purpose of asymptotic notation in algorithm analysis?

- A) To measure the exact running time of an algorithm
- B) To compare the running times of different algorithms
- C) To describe the growth rate of an algorithm's running time
- D) To determine the space complexity of an algorithm

6. Which of the following statements is true about Big Θ notation?

- A) It represents the upper bound of an algorithm's running time
- B) It represents the lower bound of an algorithm's running time
- C) It represents both the upper and lower bounds of an algorithm's running time
- D) It represents the average-case running time of an algorithm

7. What is the difference between Big O and Big Ω notation?

- A) Big O represents the upper bound, while Big Ω represents the lower bound
- B) Big O represents the lower bound, while Big Ω represents the upper bound
- C) Big O represents the average-case running time, while Big Ω represents the worst-case running time
- D) Big O represents the worst-case running time, while Big Ω represents the average-case running time

8. Which of the following functions has a time complexity of $O(n \log n)$?



- A) Binary search
 - B) Merge sort
 - C) Quick sort
 - D) Bubble sort
9. What is the time complexity of the following algorithm: for (i = 1 to n) { for (j = 1 to n) { ... } }
- A) $O(n)$
 - B) $O(n \log n)$
 - C) $O(n^2)$
 - D) $O(n^3)$
10. Which of the following statements is true about the time complexity of an algorithm?
- A) The time complexity of an algorithm is always a constant
 - B) The time complexity of an algorithm is always a linear function of the input size
 - C) The time complexity of an algorithm can be expressed as a function of the input size
 - D) The time complexity of an algorithm is always a quadratic function of the input size
11. What is the purpose of using asymptotic notation to analyze the time complexity of an algorithm?
- A) To determine the exact running time of the algorithm
 - B) To compare the running times of different algorithms
 - C) To describe the growth rate of the algorithm's running time
 - D) To determine the space complexity of the algorithm
12. Which of the following is an example of a function with a time complexity of $O(1)$?
- A) A function that performs a constant number of operations
 - B) A function that performs a linear number of operations
 - C) A function that performs a quadratic number of operations
 - D) A function that performs an exponential number of operations
13. Which phase of the System Life Cycle involves the actual construction of the system?
- A) Analysis Phase
 - B) Design Phase
 - C) Implementation Phase
 - D) Maintenance Phase
14. What is the purpose of the Analysis Phase in the System Life Cycle?
- A) To identify the system requirements and define the problem
 - B) To design the system architecture and components
 - C) To develop the system components and test the system
 - D) To maintain the system and fix errors
15. Which phase of the System Life Cycle involves the ongoing maintenance and support of the system?
- A) Planning Phase
 - B) Analysis Phase
 - C) Design Phase
 - D) Maintenance Phase
16. What is the primary output of the Design Phase in the System Life Cycle?
- A) System requirements document
 - B) System design document
 - C) System implementation plan
 - D) System testing plan



17. What is the time complexity of the following algorithm?

```
for (i = 1; i <= n; i++) {  
    // some constant time operation  
    if (i % 2 == 0) {  
        // some constant time operation  
    }  
}
```

- a) $O(1)$
- b) $O(\log n)$
- c) $O(n)$
- d) $O(n \log n)$

18. Which of the asymptotic notation is used to provide the lower bound constraint?

- A) x
- B) Ω
- C) Θ
- D) None of the above

19. Using asymptotic analysis, we can very well conclude the _____ scenario of an algorithm.

- A) best case
- B) average case
- C) worst case
- D) best case, average case, and worst case

20. Which case indicate the minimum time required for program execution?

- A) best case
- B) average case
- C) worst case
- D) None of the above

21. In a circular queue implemented with an array, how do you determine if the queue is full?

- A) $(\text{rear} == \text{front})$
- B) $(\text{rear} + 1) \% \text{size} == \text{front}$
- C) $(\text{rear} - \text{front}) == \text{size}$
- D) $(\text{front} + \text{size}) \% \text{rear} == 1$

22. What can be said about the array representation of a circular queue when it contains only one element?

- A) $\text{Front} = \text{rear}$
- B) $\text{front} = \text{rear} + 1$
- C) $\text{front} = \text{rear} - 1$
- D) None of these

23. The prefix expression $+ A * - B C D$ is given. If $A = 5$, $B = 4$, $C = 2$, and $D = 3$, what is the result of the expression?

- A) 13
- B) 7
- C) 11
- D) 15



24. The data structure that is used to implement recursion is
- A) Stack
 - B) Queue
 - C) Binary Tree
 - D) All the above
25. Convert the following infix expression into its equivalent postfix expression
 $(A+B*C)+(E^F)$
- A) $ABC*EF^{++}$
 - B) $ABC+*EF^{+}$
 - C) $AB+C*EF^{+}$
 - D) $ABC^{+}+EF^{+}$
26. Queues serve major role in _____
- A) Simulation of recursion
 - B) Simulation of arbitrary linked list
 - C) simulation of limited resource allocation
 - D) Simulation of heap sort
27. What are the disadvantages of arrays?
- a) Index value of an array can be negative
 - b) Elements are sequentially accessed
 - c) Data structure like queue or stack cannot be implemented
 - d) There are chances of wastage of memory space if elements inserted in an array are lesser than the allocated size
28. Which data structure is used for implementing recursion?
- a) Stack
 - b) Queue
 - c) List
 - d) Array
29. Which of the following is not the application of stack?
- a) Data Transfer between two asynchronous process
 - b) Compiler Syntax Analyzer
 - c) Tracking of local variables at run time
 - d) A parentheses balancing program
30. Which data structure is needed to convert infix notation to postfix notation?
- a) Tree
 - b) Branch
 - c) Stack
 - d) Queue
31. What is the value of the postfix expression $6\ 3\ 2\ 4\ +\ -\ *?$
- a) 74
 - b) -18
 - c) 22
 - d) 40
32. What data structure would you mostly likely see in non recursive implementation of a recursive algorithm?
- a) Stack
 - b) Linked List
 - c) Tree
 - d) Queue
33. Which of the following statement(s) about stack data structure is/are NOT correct?
- a) Top of the Stack always contain the new node



- b) Stack is the FIFO data structure
- c) Null link is present in the last node at the bottom of the stack
- d) Linked List are used for implementing Stacks
- 34. The data structure required for Breadth First Traversal on a graph is?
 - a) Array
 - b) Stack
 - c) Tree
 - d) Queue
- 35. The prefix form of $A-B/(C * D ^ E)$ is?
 - a) $-A/B * C ^ DE$
 - b) $-A/BC * ^ DE$
 - c) $-ABCD * ^ DE$
 - d) $- / * ^ ACBDE$
- 36. Which data structure is based on the Last In First Out (LIFO) principle?
 - a) Tree
 - b) Linked List
 - c) Stack
 - d) Queue
- 37. Which of the following application makes use of a circular linked list?
 - a) Recursive function calls
 - b) Undo operation in a text editor
 - c) Implement Hash Tables
 - d) Allocating CPU to resources
- 38. Which of the following is not the type of queue?
 - a) Priority queue
 - b) Circular queue
 - c) Single ended queue
 - d) Ordinary queue
- 39. Which of the following data structures can be used for parentheses matching?
 - a) n-ary tree
 - b) queue
 - c) priority queue
 - d) stack
- 40. Which algorithm is used in the top tree data structure?
 - a) Backtracking
 - b) Divide and Conquer
 - c) Branch
 - d) Greedy
- 41. If we want to find the last node of a linked list, then the correct coding is
 - A) `if(temp->link!=NULL)`
 `temp=temp->link`
 - B) `if(temp->data==num)`
 `temp=temp->link`
 - C) `while(temp->link!=NULL)`
 `temp=temp->link`
 - D) `while(temp->link!=data)`
 `temp=temp->link`
- 42. A linear collection of data elements where the linear node is given by means of pointer is called?
 - a) Linked list
 - b) Node list



- c) Primitive list
d) Unordered list
43. In linked list each node contains a minimum of two fields. One field is data field to store the data second field is?
- a) Pointer to character
b) Pointer to integer
c) Pointer to node
d) Node
44. The concatenation of two lists can be performed in $O(1)$ time. Which of the following variation of the linked list can be used?
- a) Singly linked list
b) Doubly linked list
c) Circular doubly linked list
d) Array implementation of list
45. Consider the following definition in c programming language.
- ```
struct node
{
 int data;
 struct node * next;
}
typedef struct node NODE;
NODE *ptr;
```
- Which of the following c code is used to create new node?
- a) `ptr = (NODE*)malloc(sizeof(NODE));`  
b) `ptr = (NODE*)malloc(NODE);`  
c) `ptr = (NODE*)malloc(sizeof(NODE*));`  
d) `ptr = (NODE)malloc(sizeof(NODE));`
46. What kind of linked list is best to answer questions like “What is the item at position n?”
- a) Singly linked list  
b) Doubly linked list  
c) Circular linked list  
d) Array implementation of linked list
47. Linked lists are not suitable for the implementation of \_\_\_\_\_
- a) Insertion sort  
b) Radix sort  
c) Polynomial manipulation  
d) Binary search
48. Linked list is considered as an example of \_\_\_\_\_ type of memory allocation.
- a) Dynamic  
b) Static  
c) Compile time  
d) Heap
49. Linked list data structure offers considerable saving in \_\_\_\_\_
- a) Computational Time  
b) Space Utilization  
c) Space Utilization and Computational Time  
d) Speed Utilization
50. Which of the following points is/are not true about Linked List data structure when it is compared with an array?



- a) Arrays have better cache locality that can make them better in terms of performance
- b) It is easy to insert and delete elements in Linked List
- c) Random access is not allowed in a typical implementation of Linked Lists
- d) Access of elements in linked list takes less time than compared to arrays

51. What does the following function do for a given Linked List with first node as head?

```
void fun1(struct node* head)
{
 if(head == NULL)
 return;
 fun1(head->next);
 printf("%d ", head->data);
}
```

- a) Prints all nodes of linked lists
- b) Prints all nodes of linked list in reverse order
- c) Prints alternate nodes of Linked List
- d) Prints alternate nodes in reverse order

52. Which of the following sorting algorithms can be used to sort a random linked list with minimum time complexity?

- a) Insertion Sort
- b) Quick Sort
- c) Heap Sort
- d) Merge Sort

53. What differentiates a circular linked list from a normal linked list?

- a) You cannot have the 'next' pointer point to null in a circular linked list
- b) It is faster to traverse the circular linked list
- c) In a circular linked list, each node points to the previous node instead of the next node
- d) Head node is known in circular linked list

54. What is the time complexity of searching for an element in a circular linked list?

- a)  $O(n)$
- b)  $O(n \log n)$
- c)  $O(1)$
- d)  $O(n^2)$

55. Which of the following application makes use of a circular linked list?

- a) Undo operation in a text editor
- b) Recursive function calls
- c) Allocating CPU to resources
- d) Implement Hash Tables

56. Which of the following is false about a circular linked list?

- a) Every node has a successor
- b) Time complexity of inserting a new node at the head of the list is  $O(1)$
- c) Time complexity for deleting the last node is  $O(n)$
- d) We can traverse the whole circular linked list by starting from any point

57. You are given pointers to first and last nodes of a singly linked list, which of the following operations are dependent on the length of the linked list?

- a) Delete the first element
- b) Insert a new element as a first element
- c) Delete the last element of the list
- d) Add a new element at the end of the list



58. In the worst case, the number of comparisons needed to search a singly linked list of length  $n$  for a given element is?

- a)  $\log_2 n$
- b)  $n/2$
- c)  $\log_2 n - 1$
- d)  $n$

59. The optimal data structure used to solve Tower of Hanoi is \_\_\_\_\_

- a) Tree
- b) Heap
- c) Priority queue
- d) Stack

60. In simple chaining, what data structure is appropriate?

- a) Doubly linked list
- b) Circular linked list
- c) Singly linked list
- d) Binary trees

61. In a binary heap, what is the time complexity of deleting the maximum element?

- A)  $O(1)$
- B)  $O(\log n)$
- C)  $O(n)$
- D)  $O(n \log n)$

62. How many edges does a complete graph with  $n$  vertices have?

- A)  $n(n-1)$
- B)  $n(n-1)/2$
- C)  $n^2$
- D)  $n^2 - n$

63. The data structure used in breadth first search algorithm is

- A) Queue
- B) Stack
- C) Heap
- D) Hash Table

64. In a max-heap with  $n$  elements, where are the leaf nodes stored?

- A) Levels 0 to  $\log n - 1$
- B) Last level only
- C) Levels  $\log n$  to  $n$
- D) Randomly

65. Which of the following traversal algorithms ensures elements are visited in sorted order for a binary search tree?

- A) Pre-order
- B) Post-order
- C) In-order
- D) Level-order

66. What is the maximum number of nodes in a binary tree of height  $h$ ?

- A)  $2^h - 1$
- B)  $2^{(h-1)} - 1$
- C)  $2^{(h+1)} - 1$
- D)  $2^h$

67. The post-order traversal of a binary tree is 8, 9, 6, 7, 4, 5, 2, 3, 1.

The in-order traversal of the same tree is 8, 6, 9, 4, 7, 2, 5, 1, 3.





The height of a tree is defined as the length of the longest path from the root to any leaf node.

What is the height of this binary tree?

- A) 1
- B) 2
- C) 3
- D) 4

68. The result of preorder traversal is same as:

- A) Depth-first order
- B) Breadth-First order
- C) Topological order
- D) Linear order

69. Which of the following is not an advantage of trees?

- a) Hierarchical structure
- b) Faster search
- c) Router algorithms
- d) Undo/Redo operations in a notepad

70. In a full binary tree if number of internal nodes is I, then number of leaves L are?

- a)  $L = 2 * I$
- b)  $L = I + 1$
- c)  $L = I - 1$
- d)  $L = 2 * I - 1$

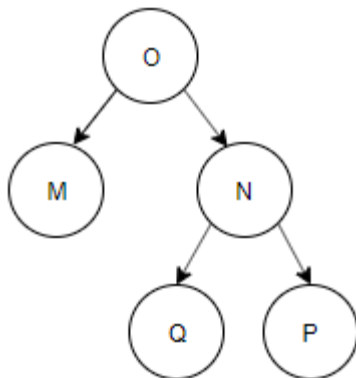
71. Which of the following is incorrect with respect to binary trees?

- a) Let T be a binary tree. For every  $k \geq 0$ , there are no more than  $2^k$  nodes in level k
- b) Let T be a binary tree with  $\lambda$  levels. Then T has no more than  $2^{\lambda-1}$  nodes
- c) Let T be a binary tree with N nodes. Then the number of levels is at least  $\text{ceil}(\log(N + 1))$
- d) Let T be a binary tree with N nodes. Then the number of levels is at least  $\text{floor}(\log(N + 1))$

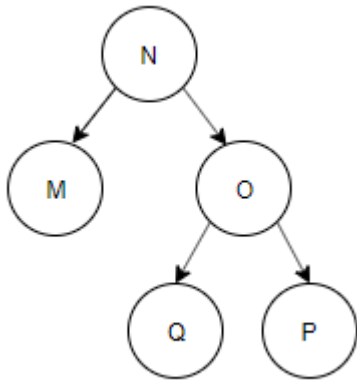
72. Construct a binary tree by using postorder and inorder sequences given below.

Inorder: N, M, P, O, Q

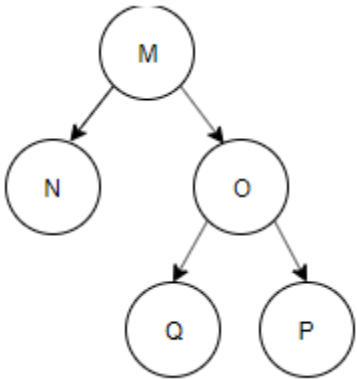
Postorder: N, P, Q, O, M



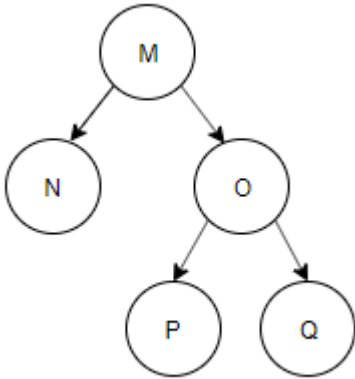
a)



b)

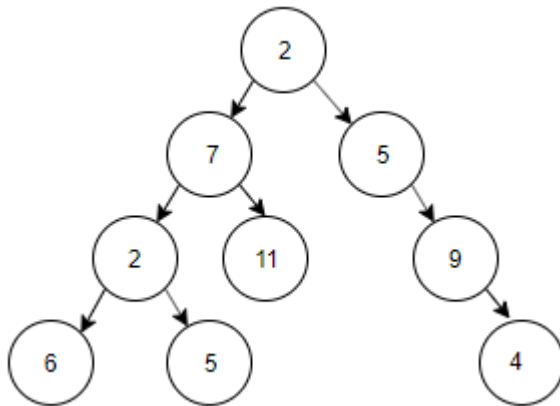


c)



d)

73. For the tree below, write the in-order traversal.



- a) 6, 2, 5, 7, 11, 2, 5, 9, 4
- b) 6, 5, 2, 11, 7, 4, 9, 5, 2
- c) 2, 7, 2, 6, 5, 11, 5, 9, 4
- d) 2, 7, 6, 5, 11, 2, 9, 5, 4

74. Which of the following graph traversals closely imitates level order traversal of a binary tree?

- a) Depth First Search
- b) Breadth First Search
- c) Depth & Breadth First Search
- d) Binary Search

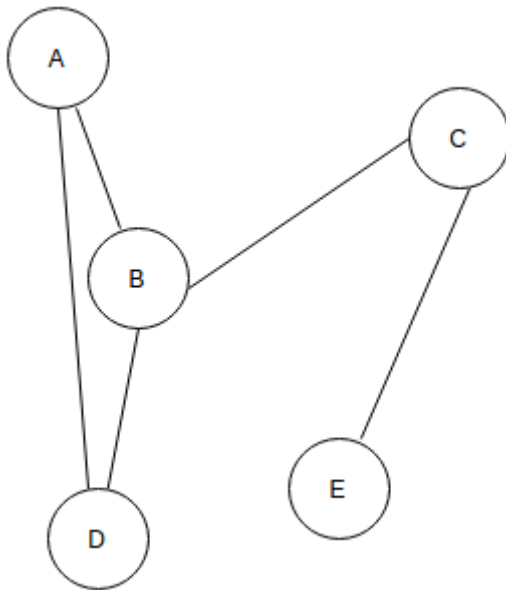
75. What is the time complexity of level order traversal?

- a)  $O(1)$
- b)  $O(n)$
- c)  $O(\log n)$
- d)  $O(n \log n)$

76. Which of the following statements for a simple graph is correct?

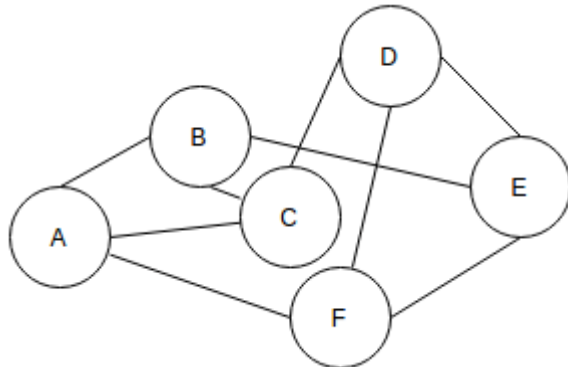
- a) Every path is a trail
- b) Every trail is a path
- c) Every trail is a path as well as every path is a trail
- d) Path and trail have no relation

77. In the given graph identify the cut vertices.



- a) B and E
- b) C and D
- c) A and E
- d) C and B

78. The given Graph is regular.



- a) True
- b) False

79. A connected planar graph having 6 vertices, 7 edges contains \_\_\_\_\_ regions.

- a) 15
- b) 3
- c) 1
- d) 11

80. Which of the following properties does a simple graph not hold?

- a) Must be connected
- b) Must be unweighted
- c) Must have no loops or multiple edges
- d) Must have no multiple edges

81. A hash table is

- A) A structure used to implement stack and queue
- B) A structure used for storage
- C) A structure that maps values to keys



- D) A structure that maps keys to values
82. A hashing function which stores colliding items together in linked lists is
- A) Separate chaining
  - B) Linear hashing with collision detection
  - C) Universal hashing
  - D) Linear hashing
83. For merging two sorted lists of size  $m$  and  $n$  into a sorted list of size  $m+n$  we required comparisons of
- A)  $O(m)$
  - B)  $O(n)$
  - C)  $O(m+n)$
  - D)  $O(\log m + \log n)$
84. If several elements are competing for the same bucket in the hash table, what is it called?
- A) Diffusion
  - B) Replication
  - C) Collision
  - D) Duplication
85. Which of the following is not an in-place sorting algorithm?
- A) Selection sort
  - B) Heap sort
  - C) Quick sort
  - D) Merge sort
86. The time complexity of heap sort in worst case is:
- A)  $O(\log n)$
  - B)  $O(n)$
  - C)  $O(n \log n)$
  - D)  $O(n^2)$
87. What is direct addressing?
- a) Distinct array position for every possible key
  - b) Fewer array positions than keys
  - c) Fewer keys than array positions
  - d) Same array position for all keys
88. What is a hash function?
- a) A function has allocated memory to keys
  - b) A function that computes the location of the key in the array
  - c) A function that creates an array
  - d) A function that computes the location of the values in the array
89. What is the load factor?
- a) Average array size
  - b) Average key size
  - c) Average chain length
  - d) Average hash table length
90. In simple chaining, what data structure is appropriate?
- a) Singly linked list
  - b) Doubly linked list
  - c) Circular linked list
  - d) Binary trees
91. Which of the following is the most widely used external memory data structure?
- a) AVL tree
  - b) B-tree



- c) Red-black tree
- d) Both AVL tree and Red-black tree
- 92. Which of the following sorting algorithms is the fastest for sorting small arrays?
  - a) Quick sort
  - b) Shell sort
  - c) Insertion sort
  - d) Heap sort
- 93. What is the advantage of selection sort over other sorting techniques?
  - a) It is faster than any other sorting technique
  - b) It is scalable
  - c) It works best for inputs which are already sorted
  - d) It requires no additional storage space
- 94. Which of the following method is used for sorting in merge sort?
  - a) partitioning
  - b) merging
  - c) exchanging
  - d) selection
- 95. Which of the following sorting algorithm does not use recursion?
  - a) bottom up merge sort
  - b) merge sort
  - c) heap sort
  - d) quick sort
- 96. Merge sort uses which of the following method to implement sorting?
  - a) selection
  - b) exchanging
  - c) merging
  - d) partitioning
- 97. Which of the following sorting algorithms is the fastest?
  - a) Merge sort
  - b) Shell sort
  - c) Insertion sort
  - d) Quick sort
- 98. Shell sort algorithm is an example of?
  - a) Bottom-up sorting
  - b) In-place sorting
  - c) Internal sorting
  - d) External sorting
- 99. In heap sort, after deleting the last minimum element, the array will contain elements in?
  - a) increasing sorting order
  - b) tree preorder
  - c) tree inorder
  - d) decreasing sorting order
- 100. Which of the following pair of sorting algorithms are stable?
  - a) gnome sort and merge sort
  - b) heap sort and merge sort
  - c) gnome sort and quick sort
  - d) merge sort and selection sort

**Answer Key**

|    |   |    |   |    |   |    |   |     |   |
|----|---|----|---|----|---|----|---|-----|---|
| 1  | B | 21 | B | 41 | C | 61 | B | 81  | D |
| 2  | C | 22 | A | 42 | A | 62 | B | 82  | A |
| 3  | C | 23 | C | 43 | C | 63 | A | 83  | C |
| 4  | A | 24 | A | 44 | C | 64 | B | 84  | C |
| 5  | C | 25 | D | 45 | A | 65 | C | 85  | D |
| 6  | C | 26 | C | 46 | D | 66 | A | 86  | C |
| 7  | A | 27 | D | 47 | D | 67 | D | 87  | A |
| 8  | B | 28 | A | 48 | A | 68 | A | 88  | B |
| 9  | C | 29 | A | 49 | C | 69 | D | 89  | C |
| 10 | C | 30 | C | 50 | D | 70 | B | 90  | B |
| 11 | C | 31 | B | 51 | B | 71 | D | 91  | B |
| 12 | A | 32 | A | 52 | D | 72 | D | 92  | C |
| 13 | C | 33 | B | 53 | A | 73 | A | 93  | D |
| 14 | A | 34 | D | 54 | A | 74 | B | 94  | B |
| 15 | D | 35 | A | 55 | C | 75 | B | 95  | A |
| 16 | B | 36 | C | 56 | B | 76 | A | 96  | C |
| 17 | C | 37 | D | 57 | C | 77 | D | 97  | D |
| 18 | B | 38 | C | 58 | D | 78 | A | 98  | C |
| 19 | D | 39 | D | 59 | D | 79 | B | 99  | D |
| 20 | A | 40 | B | 60 | A | 80 | A | 100 | A |